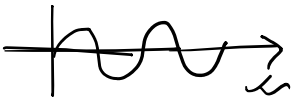
Inverter

Tuesday, April 28, 2026 9:49 PM

DC signal → (Inverter) → AC signal



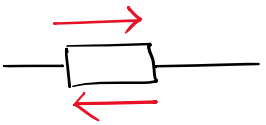


constant voltage → VFI

constant current → CFI

voltage controllable → Variable DC linked Inverter.

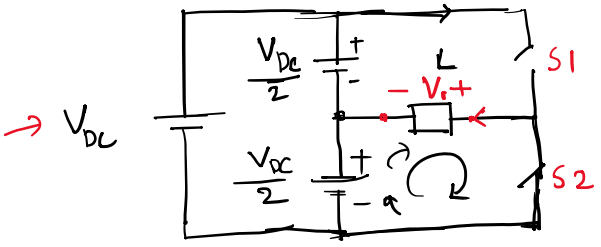
Resonant Inverter



##



S1 → ON, S2 → OFF



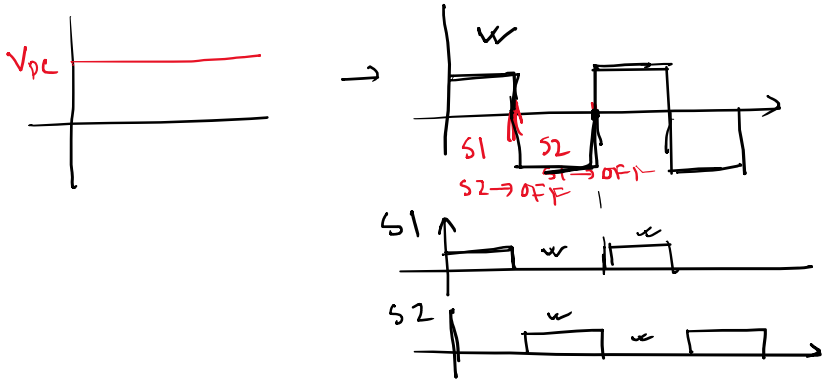
$$-\frac{V_{DC}}{2} + V_o = 0$$

$$\Rightarrow \boxed{V_o = \frac{V_{DC}}{2}}$$

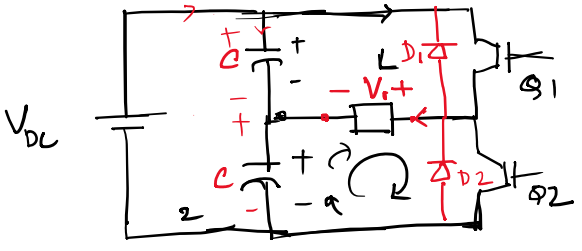
S1 → OFF S2 → ON

$$-\frac{V_{DC}}{2} - V_o = 0$$

$$\Rightarrow V_o = -\frac{V_{DC}}{2}$$

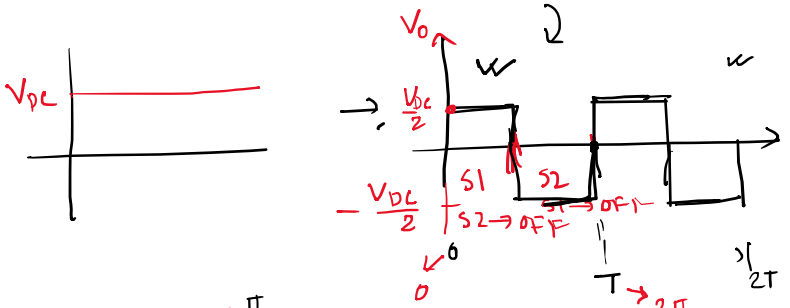


1φ half bridge Inverter.



R →

#



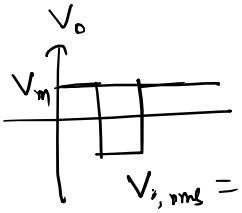
$$V_{o,rms}^2 = \frac{1}{T}\int_0^T v_o^2\,dt$$

$$V_{o,rms}^2 = \frac{1}{2\pi}\left[\int_0^{\pi}\left(\frac{V_{DC}}{2}\right)^2 d\theta + \int_{\pi}^{2\pi}\left(-\frac{V_{DC}}{2}\right)^2 d\theta\right]$$

$$= \frac{1}{2\pi}\left[\frac{V_{DC}^2}{4}\cdot\pi + \frac{V_{DC}^2}{4}\cdot\pi\right]$$

$$V_{o,rms}^2 = \frac{1}{2\pi}\cdot\frac{V_{DC}^2}{4}\cdot 2\pi = \frac{V_{DC}^2}{4}$$

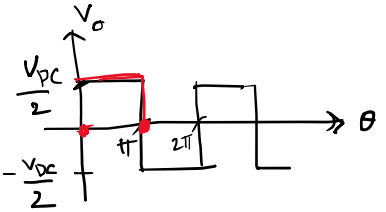
$$\Rightarrow V_{o,rms} = \frac{V_{DC}}{2}$$



$$V_{o,rms} = V_m = V_m$$

##

Odd Signal



$$V_o = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$$

$a_0, a_n \longrightarrow 0$

$$b_n = \frac{1}{T/4}\int_0^{T/2} x(t)\sin(n\omega t)\,d(\omega t)$$

$$b_n = \frac{1}{2\pi/4}\int_0^{\pi} \frac{V_{DC}}{2}\sin(n\theta)\,d\theta$$

$$= \frac{2}{\pi}\cdot\frac{V_{DC}}{2}\cdot\int_0^{\pi}\sin(n\theta)\,d\theta$$

$$= \frac{V_{DC}}{\pi}\cdot\left[-\frac{\cos n\theta}{n}\right]_0^{\pi}$$

$$= \frac{V_{DC}}{n\pi}\left(\cos 0 - \cos n\pi\right)$$

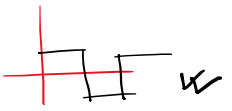
$$b_n = \frac{V_{DC}}{n\pi}\left[1 - (-1)^n\right]$$

n = even, $b_n = \frac{V_{DC}}{n\pi}(1-1) = 0$

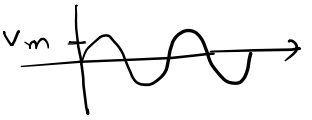
n = odd, $b_n = \frac{V_{DC}}{n\pi}(1+1) = \frac{2V_{DC}}{n\pi}$

$$V_o = \sum_{n=\text{odd}} \frac{2V_{DC}}{n\pi}\sin(n\omega t)$$

$$= \begin{cases} \sum_{n=1,3,5}^{\infty} \frac{2V_{DC}}{n\pi}\cdot\sin(n\omega t) \\ 0 \quad ; \quad n = 2,4,6,\ldots \text{ even} \end{cases}$$



$n=1$ $\quad V_{o,1} = \frac{2V_{DC}}{\pi}\sin(\omega t)$



$V_{o1} = V_{o,1\,rms} = \frac{2V_{DC}}{\pi\sqrt{2}} = 0.45\,V_{DC}$

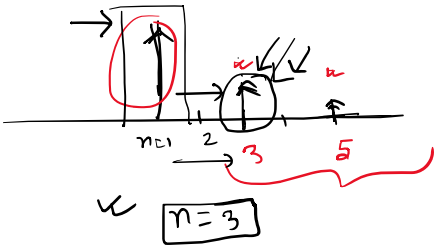
$V_{o,rms} = \frac{V_m}{\sqrt{2}}$

Harmonic Factor:-

$n \rightarrow V_{on}$

$1 \rightarrow V_{o1}$

$$HF = \frac{V_{on}}{V_{o1}}$$



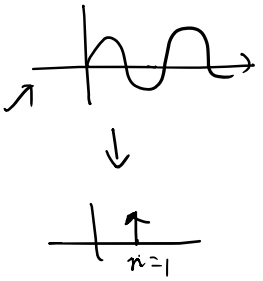
$n=3$

THD:

$$THD = \frac{1}{V_{o1}}\sqrt{\sum_{n=2,3,4}^{\infty} V_{on}^2}$$

2, 3, 4, 5 . . .

$V_h = 0$



$$= \frac{1}{V_{o1}}\sqrt{V_o^2 - V_{o1}^2}$$

$V_o \rightarrow V_{o,rms}$

$V_{o1} \rightarrow V_{o1,rms}$

$THD = 0$

$THD = \uparrow$ Distortion $\Uparrow$

Distortion Factor

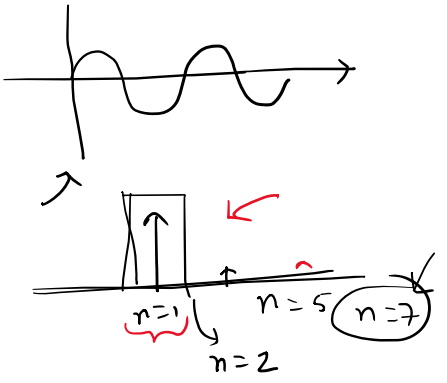
$$DF_n = \frac{V_{on}}{V_{o1}\cdot n^2}$$

$$\sqrt{\left(\frac{V_{on}}{\quad}\right)^2}$$

$$DF = \frac{1}{V_{01}} \sqrt{\sum_{n=2,3,4} \left(\frac{V_{0n}}{n^2}\right)^2}$$

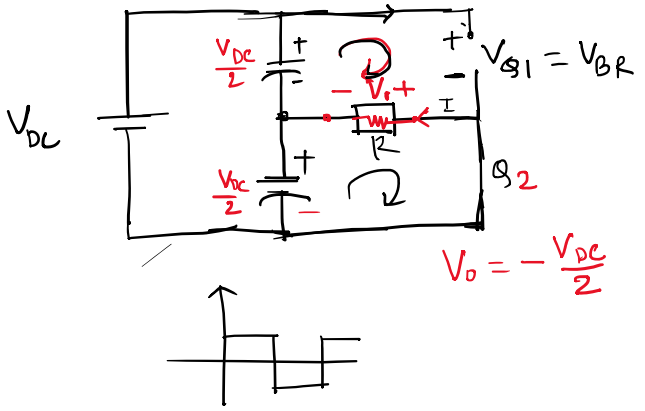
LOH:

1) frequency → closest to fundamental
2) Amplitude ≥ 3% of fundamental



$$-\frac{V_{DC}}{2} + V_{BR} + V_o = 0$$

Example: $V_{DC} = 48V$, $R = 2.4\Omega$



V_{01}

$$V_{01} = \frac{2V_{DC}}{\pi\sqrt{2}} = \frac{2 \times 48}{\pi\sqrt{2}}$$

P_o

$$P_o = I_{o,rms}^2 \cdot R = \frac{V_{o,rms}^2}{R} = \frac{24^2}{2.4} = \frac{24}{2.4} \times 24 = 240\,W$$

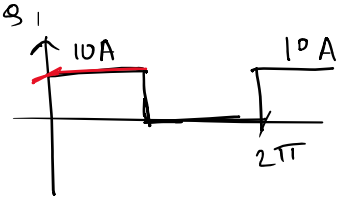
$$V_{o,pp} = \frac{V_{DC}}{2} = 24$$

$Q_1 \rightarrow ON$ $Q_2 \rightarrow OFF$

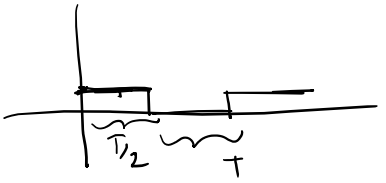
$$-\frac{V_{DC}}{2} + I \times 24 = 0 \Rightarrow I = \frac{V_{DC}/2}{24} = \frac{24}{24} = 10A$$

I_p

$I_p = 10A$



$$I_{avg} = \frac{1}{T}\int_0^T I\,dt = \frac{1}{2\pi}\int_0^{\pi} 10A\,d\theta = \frac{1}{2\pi} \times 10A \times \pi = 5A$$



V_{BR}

$$-\frac{V_{DC}}{2} + V_{BR} + V_o = 0 \qquad V_o = -\frac{V_{DC}}{2}$$

$$\Rightarrow V_{BR} = \frac{V_{DC}}{2} - V_o$$

$$= \frac{V_{DC}}{2} + \frac{V_{DC}}{2}$$

$$= V_{DC} = 48V$$

$I_{s,avg}$ $\quad P_{in} = P_o$

$$\Rightarrow I_s V_{DC} = 240W$$

$$\Rightarrow I_s = \frac{240}{48} = 5A$$

THD

$$THD = \frac{1}{V_{01}} \sqrt{V_o^2 - V_{01}^2}$$

$$= \frac{1}{0.45 V_{DC}} \sqrt{(0.5 V_{DC})^2 - (0.45 V_{DC})^2}$$

$$= \frac{0.218}{0.45} = 48.34\%$$

$$V_o = \frac{48}{2} = 24$$

$$V_{01} = \frac{2 \times 48}{\pi \sqrt{2}} = 21.6$$

$$= 0.45\, V_{DC}$$

DF

$$DF = \frac{1}{V_{01}} \sqrt{\sum_{n=2,3,4} \left(\frac{V_{on}}{n^2}\right)^2}$$

$$V_{on} = \frac{2V_{DC}}{n\pi} \sin(n\omega t)$$

$n \rightarrow$

$$\sqrt{\sum_{n=3,5,7} \left(\frac{V_{on}}{n^2}\right)^2} = \left(\frac{V_{03}}{3^2}\right)^2 + \left(\frac{V_{07}}{7^2}\right)^2 + \left(\frac{V_{05}}{5^2}\right)^2 + \ldots$$

$$= 0.024\, V_{DC}$$

$$DF = \frac{0.024 V_{DC}}{0.45 V_{DC}} = 5.38\%$$

LOH

LOH = 3 or

$$V_{on} = \frac{2V_{DC}}{n\pi}$$



$$V_{01} = \frac{2V_{DC}}{1.\pi}$$

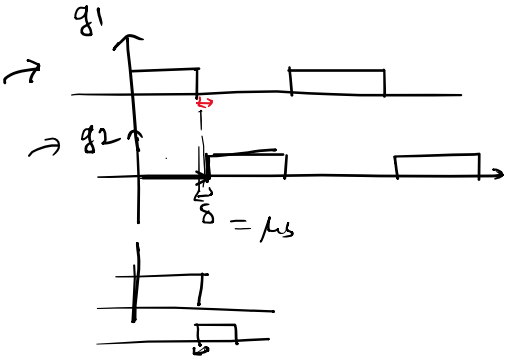
$$V_{03} = \frac{2V_{DC}}{3\pi}$$

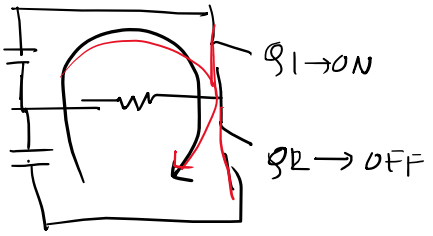
$$\frac{V_{03}}{V_{01}} = \frac{\frac{2V_{DC}}{3\pi}}{\frac{2V_{DC}}{\pi}} = \frac{1}{3}$$

$V_{03} = 33.3\% \cdot V_{01}$

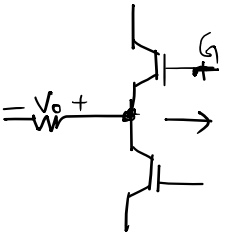
$V_{03} \geq 33\% V_{01}$

Grating sequence



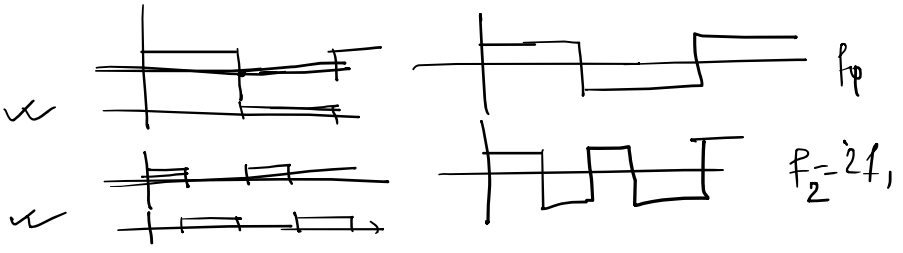


#

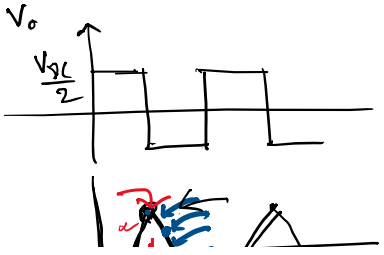


Optocoupler
Transformer
Capacitor

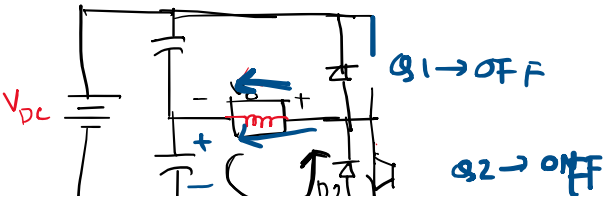
#

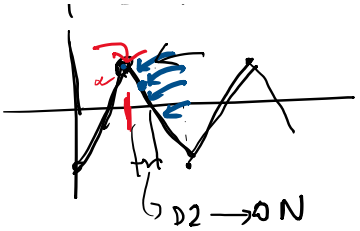


#

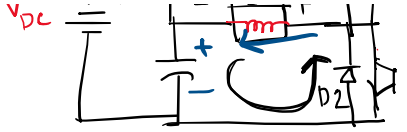


$V = 2\frac{di}{dt}$





Feedback – diodes.



IGBT

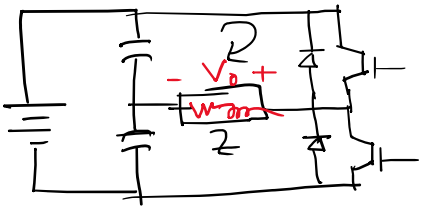
D2 → ON

Load → RL

$$V_0 = \frac{2V_{DC}}{n\pi}\sin(n\omega t)$$

$$I_0 = \frac{V_0}{Z}$$

$$= \frac{2V_{DC}}{n\pi\sqrt{R^2+(n\omega L)^2}}\cdot\sin(n\omega t - \theta_n)$$



$$Z = R + j(n\omega L)$$

$$= \sqrt{R^2+(n\omega L)^2}\angle\tan^{-1}\left(\frac{n\omega L}{R}\right)$$

$$= \sqrt{R^2+(n\omega L)^2}\angle\theta_n$$

P_{01} = $I_{0,1\,rms}^2 \times R$ $\quad n=1$

$$= \left[\frac{2V_{DC}}{\sqrt{2}\pi\sqrt{R^2+(\omega L)^2}}\right]^2 \times R$$

$$I_{0,1} = \frac{2V_{DC}}{\sqrt{2}\pi\sqrt{R^2+(\omega L)^2}}$$